Interlaboratory Comparison of Radiated Emissions in Automotive EMC

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***Abstract*—Interlaboratory comparison is performed to ensure repeatability amongst EMC laboratories worldwide regarding specific test methods. The purpose of this paper is to present an interlaboratory comparison method for radiated emissions (RE) testing used by 5 laboratories. Two statistical analysis methods are used, statistical analysis based on the ζ performance and analysis based on the z performance. A comb generator with two different antennas (to obtain broadband frequency) is used as an artefact. The sample is moved from one laboratory to another in order to obtain the raw data for the final analysis. Final report is distributed to all the ILC participants.**

Keywords—radiated emissions (RE); automotive; comb generator; ; SemiAnechoic Chamber.

I. INTRODUCTION

Participation in interlaboratory comparisons (ILCs) is an important activity that EMC laboratories should plan and perform in order to assure the quality and repeatability of tests [1], [2].

Several authors tackle the subject of interlaboratory comparison [1], [3] – [6]. Interlaboratory comparison is made to ensure that test laboratories generate technically valid data.

In [3] the authors undergo an interlaboratory comparison of radiated emissions measurement using a spherical dipole, as a reference emitter, between 30-1000 MHz in 12 OATS (Open Area Test Sites), at a 10 meter distance [3]. In India, by using a Comparison Noise Emitter developed by the University of York, interlaboratory testes were made, in the frequency range 30 – 1000 MHz, between 5 different laboratories by using the Z score [4].

Carrobi et al. carried out a proficiency test through interlaboratory comparison of radiated emission measurements. Nineteen test houses took part in the exercise providing 91 measurement results in total [1]. Measurements were performed in anechoic chambers in the frequency range comprised between 200 and 3000 MHz and the results were provided by the participants in terms of best estimate and uncertainty [1].

Lafon et al. expose the measurement results and analysis over 13 labs as well as the correlation with 3D simulation to justify about the influence of parameters individually considered (reference plane size, the absorbers performance and the grounding connection). [5].

In [6], these differences are evaluated through an interlaboratory comparison across 17 laboratories, with a special focus on the 30-100 MHz frequency range. The aim of the authors is to identify what the parameters (corresponding to degrees of freedom in the standard) leading to these observations are. [6].

In this paper, an interlaboratory method is described and the results obtained in five different laboratories are presented and commented. In Section II, the statistical analysis methods are described. In Section III, we present the measurement procedure with antenna based on a given setup. Section IV presents commented results obtained in different laboratories. Conclusions are drawn in last Section.

II. DESCRIPTION OF METHOD

The interlaboratory comparison (ILC) consists in the comparison of the measurements of a travelling Sample provided by the Coordinator of the ILC. Each Laboratory makes a quantitative measurement of the Sample in order to obtain the raw data that will be later analyzed by specific methods. The results provided by each Laboratory consist of a measured value and its uncertainty, corresponding to a given frequency.

The measurement result provided by each Laboratory was compared against: the reference value assigned by the Coordinator and the average of the measurement results provided by all participating laboratories. The Sample was represented by an electromagnetic field source made of a combination between a comb generator and an antenna. The measurement result provided by the Laboratory had to be: expressed in dB(μV/m), of the amplitude corresponding to the selected frequency; uncertainty of the estimate x, U_{lab}, expressed in dB and obtained multiplying the standard uncertainty by the coverage factor $k = 2$ (assuming normal distribution).

Two distinct statistical analyses were adopted based on two corresponding performance statistics [7]: Statistical analysis based on the ζ performance statistic: the measurement result x_i, in dB(μV/m), provided by the i-th Laboratory

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($i = 1, 2, ..., p$, where p is the number of participating Laboratories) is compared with the value X, in dB(μV/m), assigned by the Coordinator. The standard uncertainty of x_i is $u_{xi} = (U_{lab})_i / 2$ where $(U_{lab})_i$, in dB, is the expanded uncertainty stated by the i-th Laboratory [7].

The standard uncertainty of X is $u_X = U_{ref} / 2$, where U_{ref}, in dB, is the expanded uncertainty obtained multiplying the standard uncertainty by a coverage factor $k = 2$ that the Coordinator assigned to the reference value X. The Coordinator calculated the following measure ζ_i of relative deviation between x_i and X:

$$\zeta_i = \frac{x_i - X}{\sqrt{u_{xi}^2 + u_X^2}} \quad (1)$$

The measurement results provided by the i-th Laboratory will produce a warning signal if, at least at one frequency, we have ζ_i less than −2 or greater than +2. The measurement results provided by the i-th Laboratory will produce an attention signal if, at least at one frequency, we have ζ_i less than −3 or greater than +3. If at all frequencies we have ζ_i greater than −2 and less than +2 then the measurement result provided by the i-th Laboratory will not give evidence of any anomaly [7].

Statistical analysis based on the z performance statistic (ISO 13528:2005) [7] the measurement result x_i, in dB(μV/m), provided by the i-th Laboratory, is compared with the average value x^*, in dB(μV/m), obtained combining the measurement results provided by all participating Laboratories. The Coordinator calculates the following measure z_i of relative deviation between x_i and x^*:

$$z_i = \frac{x_i - x^*}{s^*} \quad (2)$$

where s^* is the estimate of the dispersion (standard deviation) of x_i about x^*, obtained combining the measurement results provided by all participating Laboratories. The value of z_i is calculated for each Laboratory and for each investigated frequency. Therefore, as many values of z_i will be calculated as the number of investigated frequencies. The measurement result provided by the i-th Laboratory will produce a warning signal if, at least at one frequency, we have less than −2 or greater than +2.

The measurement result provided by the i-th Laboratory will produce an attention signal if, at least at one frequency, we have z_i less than −3 or greater than +3. If at all frequencies we have z_i greater than −2 and less than +2 then the measurement result provided by the i-th Laboratory will not give evidence of any anomaly [7].

III. MEASUREMENT PROCEDURE

Electromagnetic field measurement must be preceded by a preliminary measurement of the power that the comb generator (CG) (Fig. 1) delivers to a 50 Ω load [8], [9]. The CG was connected to the input of a receiver through an adaptor provided by the Coordinator. It was up to the Laboratory to verify that the operation of the receiver is linear (negligible compression and generation of harmonics).

Next the deviations $\Delta = P_m - P_{ref}$, in dB and for each frequency, from the reference values P_{ref}, in dBm, were recorded. It must result that $-2,5\,\text{dB} \le \Delta \le 2,5\,\text{dB}$. If this preliminary verification produces a positive outcome at each of the frequencies, then the Laboratory can proceed to the measurement.

The critical value of 2,5 dB was derived assuming that the Laboratory uses, for both this verification and the measurement of the electromagnetic field, a meter which complies with the requirements of the standard CISPR 16-1-1:2010. It is expected that the preliminary test is passed also in case of use of a good quality, but not necessarily CISPR 25: 2008 compliant, spectrum analyzer [10].

Once that the preliminary CG power verification was completed, the electromagnetic field measurement could be carried out. The measurand is the maximum electric field strength, in dB(μV/m), emitted by the Sample in vertical polarization (±2 deg), at the specified horizontal distance from the Sample (1 m).

The measurements were performed in the frequency range 1MHz up to 1000MHz. Measuring antennas were set only in vertical polarization and at a distance of 1 meter from the reference point/antenna of the CG. The height of the test antenna was 1 meter above the ground plane of the test room. For the frequency range 1 MHz-30 MHz a rod antenna (Fig. 2) was used for measurements. For the frequency range 30 MHz-200 MHz a biconical antenna was used for measurements and for the frequency range 200 MHz-1000 MHz a log periodic antenna.

Due to the broad frequency range, two antennas are supplied with the CG-515, a 12" antenna for low frequency up to 450 MHz, and 5" antenna for 450 MHz and above. The radiated signals are generated by connecting one of the two antennas to the BNC connector located at the center of the CG-515. The circular chassis of Comb Generator helps radiate the signal more uniformly in all directions within the same plane. The conducted reference signals can be obtained by connecting a Coaxial Cable between the BNC connector mentioned above and a Spectrum Analyzer.

Step size at the output of the comb generator was set according to the following: 1-30MHz with step 1 MHz and 30-1000MHz with step of 5 MHz. Resolution bandwidth and measuring time at the test receiver were: 9 kHz for 1MHz-30MHz (50 ms), and 120kHz for 30MHz-1000MHz (10 ms).



Fig. 1 CG515 Antennas

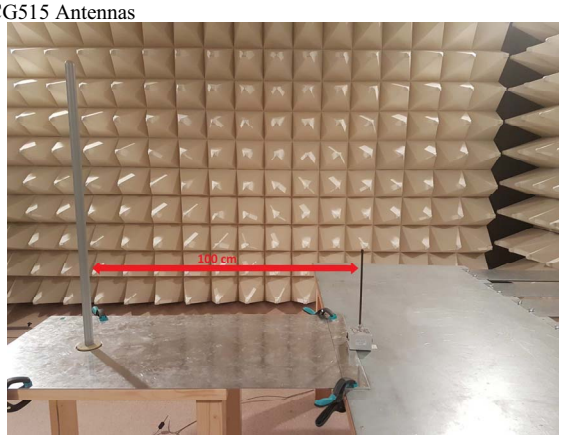


Fig. 2 Rod Antenna 1-30 MHz

IV. MEASUREMENT RESULTS

For the Interlaboratory Comparison Radiated Emissions in the 1 - 1000 MHz frequency range - ILC scheme, five laboratories have taken part in measurements and were considered in the final report. In order to keep confidential any information pertaining the performance of the Laboratories involved in the ILC test during its progress and after its completion, the coordinator named these laboratories as follow: Reference Lab, Lab 1, Lab 2, Lab 3, Lab 4, and Lab 5.

All data reported below will refer to the scoring system described above, in order to keep anonymous the results associated with each laboratory. Next, the recordings and final results will be presented for all five laboratories involved in the ILC scheme for RE.

The measured electric field strength x, in dB(μV/m), was rounded up to 1 decimal. The values of x and U_{lab} were recorded by each Laboratory individually for several frequencies (1MHz, 5-30MHz (step 5 MHz), 30-50MHz (step 10MHz), and 100MHz -1000MHz (step 100MHz)). The coordinator recorded: X, U_{ref}, x*, s*, x-X, x-x*, ζ and Z score for the same frequencies.

In Fig. 3 we can see the measured electric field values (x) for all five laboratories. Between 20 and 50 MHz significant differences can be noticed due to the differences in the Semi-Anechoic Chambers (SAC) [11]. At 20 MHz, Lab 5 has a measured value of approximatively 50 dBuV/m, whilst Laboratories 1-4 have aprox. 34 dBuV/m. At 30 MHz, different results are also obtained. Whilst Laboratory 5 has a measured value of 25 dBuV/m, Laboratory 3 has 34, and Laboratory 1 has 44.

In Fig. 4 the statistical analysis based on ζ performance was presented. We can see that Lab 5 has different results between 10 and 40 MHz. Also at 20 MHz, Lab 5 exceeds the value 2, and at 30 MHz, Lab 4 exceeds the value 2. These values produced warning signals for the two laboratories, by taking into consideration the procedure described in chapter two [7].

In Fig. 5 we can see the Z score for all five laboratories. Only Laboratory 5 exceeds the upper limit (+3) of the Z score, so an attention signal was sent to this laboratory as described in chapter 2 [7].

Probable reasons for obtaining different results between 10-100MHz could be: large measurement uncertainty declared by Laboratory 5 (Table 3), the difference in the size of the chamber, ground plane size, ground connections, the absorbers performance due to the difference in geometry and material types, different types of antennas, mismatch and performance of the receiver and also cable corrections [4], [5]. The deviations form the above described frequencies result from the high VSWR of the receiving antenna at the corresponding frequencies. [3], [4].

Table 1 ±Uncertainty for the involved laboratories (at 30 MHz)

Laboratories	$+U_{lab}$ [dB]	$-U_{lab}$ [dB]
Lab 1	2	-3
Lab 2	3.9	-3
Lab 3	3.5	-7.2
Lab 4	3.98	-3.98
Lab 5	6.9	-13.5

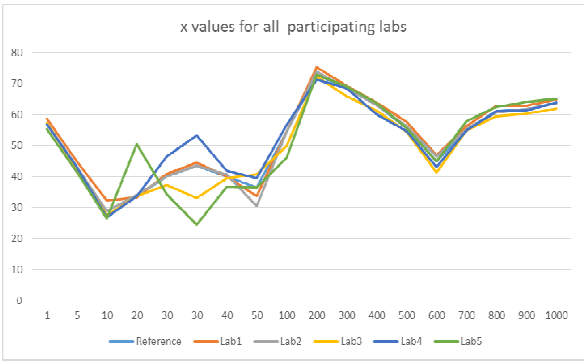


Fig. 3 X values

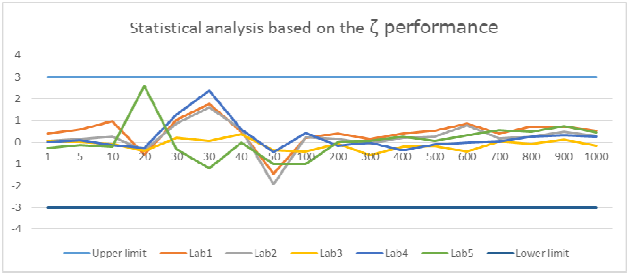


Fig. 4 Statistical analysis

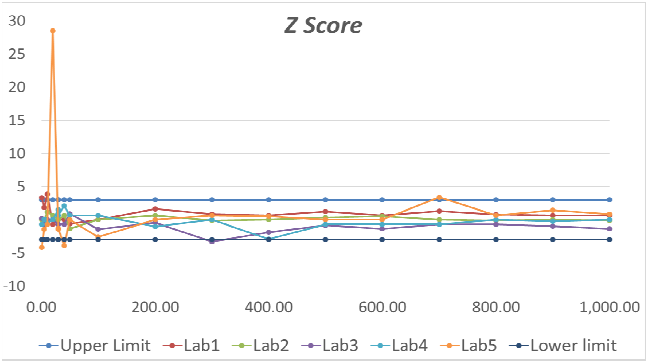


Fig. 5 Z score

The antenna factor (AF) of an antenna varies with the antenna height because of mutual coupling with the ground plane. This effect on RE measurements is not specified by manufacturers and also it is not mentioned in the calibration certificates [3], [5].

V. CONCLUSIONS

In this paper we reported the use of an artefact for interlaboratory RE testing in 5 different laboratories. Two statistical analysis methods are used, statistical analysis based on the ζ performance and analysis based on the z performance. A comb generator CG-515 with two different antennas (to obtain broadband frequency) was used as an artefact. The measurements were performed in the frequency range 1MHz - 1000MH with vertically polarized antennas: rod, biconical, and logperiodic. The sample was moved from one laboratory to another in order to obtain the raw data for the final analysis.

By studying the measured electric fields from all 5 laboratories we can see that significant differences appear only between 20 and 50 MHz. The results from the ζ performance show that Lab 4 and 5 exceed the value +2, so warning signals are sent to those laboratories. Regarding analysis based on the z performance, Lab 5 exceeded the upper limit (+3) so an attention signal was sent to this laboratory.

A cause may come from large measurement uncertainty declared by this laboratory, from the cables used or from the rod antenna. Based on this value that exceeded the upper limit, a non-conformity report was started with a few corrective actions to be implemented: a new measurement according to the ILC plan will be taken for Lab 5 and based on this new measurement a new statistical analysis will be calculated. This new results will be part of a future work which will include also in a new ILC program for conducted emission (CE) test method.

Because the z score is highly influenced by the number of the involved laboratories, as part of a future work the authors plan to extend the number of laboratories in a new ILC plan consisting in at least two quantitative test methods (at least RE and CE).

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